

Depth of Processing Approach to Face Recognition: A Test of Two Theories

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Three experiments investigated feature quantity and semantic quality accounts for level of processing effects on face recognition. Experiment 1 established that as the "level of processing" of a face judgment increased, the number of eye movements and inspection time increased, and subsequent recognition performance (recognition and recognition response time) improved. Experiment 2 found that, with the number of eye movements and inspection time held constant, as the "level of processing" of a face judgment increased subjects' cognitive level of processing (assessed by task-evoked peak pupillary dilation) increased, but recognition performance did not improve. Experiment 3 indicated that, with the number of eye movements held constant, recognition performance did not improve as a function of inspection time. These findings are consistent with a **feature quantity hypothesis**.

It is well established that performance during face recognition is influenced by the instructions given at the time of initial inspection. For example, when subjects are asked to make personality trait judgments (e.g., "niceness") at the time of initial inspection of unfamiliar faces, they have significantly better recognition performance than when they are originally asked to make judgments about physical characteristics such as gender (e.g., Bower & Karlin, 1974, Experiments 1 and 2). This difference in recognition performance has come to be called the "levels of processing effect," an effect replicated in a number of studies (e.g., Mueller, Courtois, & Bailis, 1981; Strnad & Mueller, 1977).

Bower and Karlin (1974) first demonstrated the levels of processing effect in face recognition and offered an explanation similar to that for previous memory studies using word stimuli (e.g., Craik, 1973; Craik & Lockhart, 1972; Craik & Tulving, 1975), which concluded that words processed for their "semantic meaning" were remembered better than words processed for their physical attributes. Bower and Karlin suggested that, comparable to semantic word meaning, faces cue "unique associations" (e.g., the face may remind subjects of someone they know or perhaps a famous person). As processing level increases, a more "semantic" type of process-

ing occurs in which an increased number of unique associations to the target face are formed. Such a mechanism, identified here as the **semantic quality hypothesis**, is thought to create a richer, more semantic network of association pathways to a memorial representation of that face and thus to enhance subsequent recognition.

An alternative explanation, one that Bower and Karlin (1974) ultimately favored, held that there may be a difference in the number of specific features viewed and encoded at study time. According to this explanation, relatively few features need to be viewed before a judgment about a physical attribute can be made, whereas a greater number of features must be viewed before a judgment about a personality trait can be made. As so-called "level of processing" increases, the number of features encoded increases and, as a result, so does subsequent recognition. This explanation is presently referred to as the **feature quantity account**.

The feature quantity account has become controversial. Although three studies (McKelvie, 1985; Patterson & Baddeley, 1977; Wells & Hryciw, 1984) suggest that the levels of processing effect are not attributable to the number of facial features encoded, five other studies (Blaney & Winograd, 1978; Courtois & Mueller, 1979; Winograd, 1976, 1978, 1981) yielded findings that lend support to Winograd's (1978) "elaboration hypothesis." The elaboration hypothesis, like the feature quantity account, contends that personality trait judgments lead to a greater number of features viewed and encoded during inspection (in addition, Walker-Smith, Gayle, & Findlay, 1977, found eye movement data that showed the importance of facial features in face recognition).

The present study was designed to shed further light on the semantic quality account versus the feature quantity account for the levels of processing effect in face recognition.

Experiment 1

In the present experiment, we set out first to replicate the findings of previous studies—that as level of processing increases, so does subsequent face recognition performance.

Experiment 1 is based on an independent study completed at Gettysburg College by Lance C. Bloom and was presented at the 61st Annual Meeting of the Eastern Psychological Association in Philadelphia, April 1, 1990.

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Second, we sought to move beyond the demonstration of a levels of processing effect to test differentially the quality versus the quantity hypotheses to account for that effect.

To test the feature quantity hypothesis, we recorded eye movements during inspection (learning), thus making it possible to observe how many features of a target face were examined. We presumed that the number of features attended to was positively correlated with the number of features actually encoded by subjects. The feature quantity hypothesis predicts that as processing level increases, the number of eye movements (features encoded) will increase, and so will later face recognition.

We used two converging measures to test the alternative (semantic quality) hypothesis—one, a physiological measure taken at the time of initial inspection (learning) of the faces and a second, psychometric measure taken during the recognition task. These measures have been shown in other contexts to be related to cognitive level of processing and cognitive level of representation, respectively (e.g., Beatty & Wagoner, 1978; Blatt et al., 1979).

The physiological measure, pupil diameter, has been shown by Beatty and Wagoner (1978) to be related to cognitive processing level in the standard Posner and Mitchell (1967) matching paradigm. Both peak magnitude and latency to peak magnitude of the task-evoked pupil response were found to increase when subjects made increasingly deep (semantic) physical, name, and category matches for letters (Beatty & Wagoner, 1978, Experiment 2).

Other information processing studies that have investigated complex cognitive processing tasks (Kahneman, Beatty, & Pollack, 1967), levels of task difficulty in digit string transformations (Kahneman, Tursky, Shapiro, & Crider, 1969), arithmetic problem solving (Hess & Polt, 1964), short-term memory for digits and words (Kahneman & Beatty, 1966; Kahneman, Onuska, & Wolman, 1968; Kahneman & Peavler, 1969), pitch discrimination (Kahneman & Beatty, 1967), and imagery (Paivio, 1973) consistently report that between different types of tasks, peak pupillary dilation increases as the subject's level of cognitive effort increases in response to the task demands. Furthermore, these studies have demonstrated that within a single trial, the moment-to-moment pupillary dilation reflects momentary variations in the processing load that the task imposes on the subject (Kahneman & Peavler, 1969).

Together, the findings of Beatty and Wagoner (1978) and others suggest that as the level of processing of a mental task increases, the cognitive effort used (or the demand for cognitive resources) increases. Collectively these studies consistently show that cognitive effort or processing load can be sensitively and reliably measured by pupillary dilation (see Beatty, 1982, for a review).

In the present study, the semantic quality hypothesis predicts that pupil diameter will be significantly greater for a trait judgment (deep processing) inspection condition than for a gender judgment (shallow processing) inspection condition.

The second measure of processing level, the psychometric measure, used in the present study was the Blatt, Chevrone, Quinlan, and Wein (1981) measure of Qualitative and Structural Dimensions of Object Representations. The Blatt instru-

ment is a projective test developed from developmental cognitive and psychoanalytic theory to assess, among other things, the conceptual representational level of significant objects (e.g., Blatt et al., 1979). The test involves a judge's rating, on a scale of 1 to 9, of subjects' open-ended responses to the query "please describe [the face that you saw]." A rating of 1 indicates a relatively primitive *sensorimotor-preoperational* level of representation, and a 9 represents an abstract *conceptual representation*. In the context of this experiment, we presumed, as in any projective task, that the subject's prior judgment experience (memory network) from initial learning would necessarily be brought to the recognition-description task so that the conceptual representational level of the object (face) could be assessed.

In the present study, the semantic quality hypothesis predicts the following differences in type of representational information retrieved from memory between processing levels: For a sex judgment (physical-external, low processing level), a score of about 3 is expected. With a score of 3 (concrete-perceptual level of representation), "Emphasis is often on what the person looks like in its external characteristics or physical properties, in a literal concrete sense" (Blatt et al., 1981, p. 14). For a trait judgment (internal personality attribute, high processing level), a score of about 7 is expected. With a score of 7 (internal iconic level of representation), "The person is described in terms of his/her attributes and. . . . The description is directed toward internal dimensions" (Blatt et al., 1981, p. 14).

The three measures relevant to the underlying basis of the processing levels effect in face recognition in this experiment can be related to the two alternative hypotheses as shown in Table 1 (upper section). If the quantity hypothesis is correct, there should be no significant difference between trait (deep processing) and gender (shallow processing) judgments on the pupillometric and Blatt measures, but there should be a difference in the number of eye movements. The opposite should be found if the quality hypothesis is correct, namely, no difference between the two processing levels for the number of saccades but significantly greater pupil size and Blatt scores for the trait judgment condition as compared to the gender judgment condition.

Method

Subjects. Eighteen Gettysburg College students (nine women and nine men) who were required to participate in several experiments to obtain credit for their introductory psychology course served as participants to satisfy one of those requirements.

Stimuli. We used nine Caucasian faces constructed from a Photofit face kit (kits PFW 100-181 and PFW 500-28). These faces were copied on a mita photocopier (model DC-2055) so that there were no artificial "lines" (e.g., Goldstein & Chance, 1981) compartmentalizing the face. All faces had neutral expressions. Five of the faces were homogeneous male faces, and four were homogeneous female faces (distinguishing characteristics, such as jewelry, glasses, facial hair, and so forth were avoided). We constructed 27 faces (the 9 inspection faces plus 18 distractors) from the Photofit kit, copied them, and photographed them into slides. Each of the original nine inspection slides had two highly similar distractor slides. Each of these nine sets of three similar slides (within sets) had identical hair, forehead, and

Table 1
Experiment 1: Predictions and Results for Three Probe Measures as a Function of Level of Processing (Face Judgment)

Hypothesis	Number of saccades		Pupil diameter	Blatt measure		
Predictions						
Feature quantity	LP < HP		LP = HP	LP = HP		
Semantic quality	LP = HP		LP < HP	LP < HP		
Results						
	<i>LP</i> ^a	<i>HP</i> ^a	<i>LP</i> ^a	<i>HP</i> ^a	<i>LP</i> ^b	<i>HP</i> ^b
<i>M</i>	4.68	< 10.96	93.93	= 93.16	4.22	= 3.92
<i>SD</i>	5.33	10.82	24.36	22.39	1.49	1.42

Note. LP represents low-processing condition; HP represents high-processing condition. < indicates a significant difference ($p < .01$); = indicates no significant difference.
^a $N = 16$. ^b $N = 18$.

ears but differed on three or four of the following features: chin, mouth, nose, eyes, and eyebrows. Across the nine sets (between sets), each of the five male face sets had similar, but not identical, hair (length, style, and to a lesser degree color), forehead, and ears. The four female face sets were constructed in the same way.

Apparatus. For the inspection task, each of the nine faces was imaged by a Panasonic video camera (model vv-cd 20), which presented the face to subjects on an AMDEX 12" black-and-white video monitor (model VIDEO 100). The monitor was approximately 25.4 cm above a table surface that stood about one meter above the floor. Subjects sat in a chair and used a fixed "bite-bar" to keep their head still and their eyes consistently 35.6 cm from the monitor. The bite-bar was approximately 26.5 cm above the surface of the table. The faces presented on the monitor covered most of the vertical display area so that the image on the monitor was approximately 12.7 cm (wide) $\times$ 16.5 cm (high). This image subtended a visual angle of approximately 20° 12' $\times$ 26° 6'.

An infra-red illuminator was placed next to the monitor, and a Pulnix black-and-white video camera (model TM 34K) with an infra-red filter was placed behind the illuminator. The Pulnix camera monitored the dark pupil relative to the reflection of infrared light from the surrounding iris, sclera, etc. of the subject's right eye, and sent the reflection data to the ISCAN Pupil Tracking System (model RK 416) and the ISCAN Autotracking System (model RK 520). Pupil diameter and eye location were registered every 16 ms.

The eye movement, pupillometric, and frame count (time) data from the ISCAN were recorded by a Panasonic VCR (model AG-630) and played back later for manual scoring. This VCR model had a 32-ms frame-by-frame advancement capability, so that length of inspection time and pupil diameter could be accurately recorded (digital) and saccade data (analogue) could be scored reliably. Despite unavoidable noise introduced into the data by small head and body movements, blinks, and so forth, we obtained scorable records for 16 subjects under all instructional conditions. Data were collected at normal illumination levels (24 fc).

For the recognition task, the 27 slides were projected onto a screen (71 $\times$ 112 cm) using a Kodak Ektographic slide projector (model E-2) and a prearranged slide carousel. The image of these faces was approximately 46 $\times$ 70 cm in size projected 2.1 meters from the subject. This image subtended a visual angle of approximately 12° 30' $\times$ 18° 54'. A millisecond timer measured subjects' recognition response time (RT) for each of the 27 slides. Finally, a response sheet prepared to conform to the Blatt et al. (1981) instructions stating, "please describe the face that you just saw," was given to subjects at

appropriate times. Relative to the inspection task, room illumination was lower during recognition (.6 fc) to facilitate slide viewing.

Design. The experiment had one (within subjects) independent variable, level of processing, defined in terms of the instructions given to subjects to make judgments about faces during the inspection task. These instructions had three levels: no (subject merely indicates when a face is "seen"), low (gender judgment), and high (honesty judgment). Subjects inspected a total of nine faces and made a no-processing judgment for three of the nine faces, a low-processing judgment for three other faces, and a high-processing judgment for the final three faces. Order of judgments was counterbalanced across subjects, but subjects always completed one level of processing (three faces) before proceeding to another level. The three faces presented at each processing level were counterbalanced, as was the serial presentation order of the three faces at each processing level. For each inspection trial, eye movement, pupil diameter, and inspection time data were recorded from stimulus onset to subject's eye closure. We took six dependent measures, three confirming measures, and three probe measures.

Confirming measures. The first three measures were designed to confirm the replication of the levels of processing effect. We expected these three confirming measures (recognition, inspection time, and recognition RT) to yield significant differences if in fact the levels of processing effect was replicated by the gender versus trait inspection instructions. Recognition and inspection time were expected to be significantly greater at the deepest level of processing under both hypotheses (semantic quality and feature quantity). Recognition RT was expected to be significantly shorter at the deepest level of processing under both hypotheses.

We determined recognition at each level of processing as the number of inspection faces correctly recognized. Inspection time was calculated by subtracting the ISCAN frame count at initial stimulus presentation from the ISCAN frame count when the subject closed his or her eyes to indicate termination of inspection. Recognition RT (of correct yes responses) was obtained with a digital millisecond timer equipped with a photoconductor "on" switch, triggered by the presentation of a recognition slide, and subject response "off" switches (yes or no).

Probe measures. The three remaining dependent measures (eye movements, pupil diameter, and Blatt scores) were designed to elucidate the mechanism underlying the replicated effect.

Eye movement scoring and reliability. The number of saccades (during inspection) was established by a 32-ms frame-by-frame analysis of the tapes that displayed a cursor representing the subject's line of sight superimposed over a representation of the target face during inspection. Any saccade approximately 10 mm (1° 36') or larger was considered to yield a new fixation point (an exception to this rule was when subjects saccaded from an eye to a brow or vice versa, because for one face the eyes and brows were as close as 30'). In most instances, the saccades were obviously discernable from noise. To check scoring reliability, a second, trained rater who was naive to the entire experiment independently scored the eye-movement data tapes. The interrater reliability was .95.

Pupil diameter scoring and reliability. Pupil diameter (during inspection) was indexed as the average number of display pixels subtended by the subjects' pupil image every 32 ms during the inspection task. This measure was obtained from a digital readout of the ISCAN RK 520.

Blatt et al. (1981) scoring and reliability. Subjects' written responses (during recognition) to the instruction "please describe the face that you just saw" were coded by a laboratory technician experienced with the Blatt measure. Scoring reliability was established by having the technician rescore the data 4 months later. The correlation between the rater's two scorings was .86, indicating a satisfactory test-retest reliability.

Procedure. Subjects were run individually and informed that they would view a number of faces about which they would have to make specific judgments. They were then positioned at a bite-bar, at which time the infrared illuminator and infrared camera were adjusted so that they were centered on the subject's right eye. A stable threshold was then obtained so that registration-calibration of the subject's eye positions to a standard inspection pattern could be performed.

During the inspection task, subjects were given the following instructions prior to each presentation of a face. In the control or no-processing condition, they were instructed, "as soon as you see the face, please simply close your eyes and keep them closed." In the low-processing condition, subjects were instructed, "please judge whether the face is male or female, and as soon as you have made your judgment, please close your eyes and keep them closed." In the high-processing condition, subjects were instructed, "please judge, using your own criteria, whether the face is honest or not honest, and as soon as you have made your judgment, please close your eyes and keep them closed." After a judgment had been made for each face (nine faces total), subjects were presented with a blank monitor and asked to open their eyes and focus on a fixation point. The experimenter then adjusted the system and prepared another face (aligning the center of the nose and the fixation point) for the next trial.

The recognition task presented subjects with 27 slides (the nine inspection faces intermixed according to a counterbalanced order with 18 distractor faces) and required subjects to indicate whether they recognized the face (by pushing a *yes* switch) or did not recognize the face (by pushing a *no* switch) from the previous inspection task. After each of the original nine inspection faces was presented, the subject performed the Blatt task.

Results and Discussion

Eye movement data, inspection time data, and pupil diameter data were lost for two subjects because of equipment malfunction. For the remaining 16 subjects, 30 (out of 48 possible) means for these three variables were calculated from all three trials (stimulus faces per condition), 10 were calculated from two trials, and 8 were based on just one trial. There was no apparent systematic pattern in the loss of these data.

For six subject cell means, the mean number of faces correctly recognized was zero (four means in the no-processing condition and two means in the low-processing condition). Thus, no recognition RT scores for correct *yes* responses were available for those subjects at those processing levels. So that the repeated measures analysis of variance ($n = 18$) could be performed for recognition RT, we calculated the two processing level means for recognition RT independently ($n = 14$ for the no-processing level mean and $n = 16$ for the low-processing level mean). Those means were inserted into the four no-processing and two low-processing blank subject cell means, respectively, for purposes of the analysis.

Confirming measures. The mean number of faces recognized, 1.44 ($SD = .98$), 1.67 ($SD = .84$), and 2.28 ($SD = .75$), increased as processing level increased, $F(2, 34) = 5.91$, $p < .01$. These means translate into hit rates of .48, .56, and .76, respectively. Newman-Keuls tests indicated no significant difference between the no- and low-processing levels, but each of those conditions differed significantly from the high-processing condition (no vs. high, $p < .01$; low vs. high, $p < .05$). The overall false positive rate was .34.

As processing level increased, so did the mean length of inspection time: .39 s ($SD = .19$) for no-processing, 1.09 s ($SD = 1.45$) for low-processing, and 2.78 s ($SD = 2.63$) for high-processing, $F(2, 30) = 10.06$, $p < .001$. Newman-Keuls tests revealed no significant difference between the no- and low-processing levels, but each of those conditions differed significantly from the high-processing condition (no/high, $p < .01$; low/high, $p < .01$).

The mean recognition RT of correct *yes* responses for the no-, low-, and high-processing levels was 3.43 ($SD = 2.11$), 3.82 ($SD = 1.71$), and 2.98 ($SD = 1.41$) s, respectively. An analysis of variance indicated no significant main effect of processing level on recognition RT, $F(2, 34) = 1.30$, $p > .05$; but a separate follow-up analysis of variance indicated a significant main effect of processing level on recognition RT between the two experimental conditions (low- vs. high-processing), $F(1, 17) = 4.99$, $p < .04$. The mean recognition RT of correct *yes* responses significantly decreased as processing level varied from low to high.

The three confirming measures (recognition, inspection time, and recognition RT) in fact confirmed that the instructions given to subjects in the trait versus gender judgment tasks induced significant differences, the levels effect (e.g., Bower & Karlin, 1974; Mueller, Carlomusto, & Goldstein, 1978; Smith & Winograd, 1978; Warrington & Ackroyd, 1975). These measures established that, because the recognition task was constant for all conditions, something in the inspection instructions led to the characteristic recognition difference, a difference in recognition RT, and a difference in inspection time during which different mechanisms presumably operated.

Probe measures. The mean number of saccades increased as processing level increased from no to low to high, 1.52 ($SD = 1.64$), 4.68 ($SD = 5.33$), and 10.96 ($SD = 10.82$), respectively, $F(2, 30) = 10.39$, $p < .001$. Newman-Keuls tests indicated no significant difference between the no- and low-processing levels, but each of those conditions differed significantly from the high-processing condition (no vs. high, $p < .01$; low vs. high, $p < .01$).¹

Mean pupil diameters for the no-, low-, and high-processing levels were 101.28 ($SD = 29.0$), 93.93 ($SD = 24.36$), and 93.16 ($SD = 22.39$), respectively. These ISCAN pixel scores translate into a range of 3.33–3.11 mm actual pupil diameter (based on the calibration of the pixel index to an artificial pupil placed 38 cm from the camera). An analysis of variance indicated no significant main effect of processing level on pupil diameter, $F(2, 30) = 2.14$, $p > .05$.

The mean scores for the Blatt measure of conceptual level of representation for the no-, low-, and high-processing levels were 3.99 ($SD = 1.36$), 4.22 ($SD = 1.49$), and 3.92 ($SD = 1.42$), respectively. Analysis of variance indicated no signifi-

¹ A saccade was made, on average, every 257, 233, and 254 ms for the no-, low-, and high-processing conditions, respectively. Within the limitations of the 32-ms video sampling rate, we estimated that saccades typically lasted from a little less than 32 ms to a little less than 64 ms, depending on the distance traveled, and that the average saccade duration was approximately 40–50 ms, with the balance of time allotted to the fixation.

cant main effect of processing level on the Blatt measure of conceptual level of representation, $F < 1$.

We examined the distribution of the Blatt scores to assess the possibility that the nonsignificant differences observed were due to the insensitivity of the instrument rather than to the (inferred) lack of conceptual level of representation differences across the three types of judgments. Given a scorer reliability of .86, which indicated the presence of substantial nonrandom variance, we presumed that obtained score variability reflected the sensitivity of the instrument. The scores obtained ranged from 3 to 7 on a 9-point scale, covering 56% of the total scale range. Furthermore, the standard deviations for the three judgment types ranged from 1.36 to 1.49, with an average of 1.42, corresponding to 15.8% of the available range (1–9) and 28.4% of the obtained range (3–7). The Blatt score distributions obtained, in conjunction with the fact that we used a within-subjects experimental design, provide some assurance that the test was assessing true differences in subjects' conceptual level of representation in the present experiment, as it has in previous work (e.g., Blatt et al., 1979; Blatt et al., 1981).

The three probe measures in the present experiment were designed to address the nature of the mechanism—whether a difference in the number of features encoded (quantitative hypothesis) or a difference in the semantic richness of the encoding (qualitative hypothesis)—underlying the replicated levels of processing effect. Table 1 (lower section) summarizes the experimental results of the probe measures in juxtaposition with the predictions specified in the upper section of the table. It is clear that the pattern of results across the three measures is consistent with the feature quantity hypothesis. Before conclusive inferences can be drawn from the pattern of results for the three probe measures, however, two considerations concerning the relation between the number of eye movements observed and two of the other measures should be addressed empirically.

The first consideration raised by the present results is the inescapable confound between inspection time and number of eye movements in a natural (subject controlled) viewing paradigm. We addressed this issue more fully in Experiment 3.

The second consideration is that the present procedure in which the pupil measure was obtained varied from that used by Beatty and Wagoner (1978). They presented pairs of letter stimuli to subjects for 100 ms, so that subjects' eyes were fixed during inspection, and then measured the pupillary response over a subsequent 2-s judgment interval. In addition, the pupil measure used in Experiment 1, average pupil diameter, differed from the Beatty and Wagoner (1978) task-evoked pupillary response measure (peak dilation and latency to peak dilation). In short, the burden rests with us to establish the comparability of the present levels of processing paradigm with that of Beatty and Wagoner (1978). Experiment 2 was designed to address that issue.

Experiment 2

Comparability of the present paradigm and the Beatty and Wagoner (1978) paradigm was established by presenting faces

to subjects for 96 ms so that they could not move their eyes during inspection. We then measured the pupillary response during the ensuing 3-s judgment interval (the 3-s judgment time was selected to accommodate the range of unrestricted mean judgment times found in the first experiment). Consequently, in addition to pupil diameter, peak pupillary dilation and latency to peak pupillary dilation were assessed in a manner directly comparable to that of Beatty and Wagoner (1978).

The question of interest in the present study was, with the length of inspection time held constant (and no eye movements allowed), would subjects' cognitive level of processing (assessed by task-evoked pupillary response) increase significantly and subsequent recognition performance improve significantly as the level of processing of a face judgment varied from "no" to "gender" to "honesty"? The semantic quality hypothesis predicts that recognition, pupil diameter, peak pupillary dilation, latency to peak pupillary dilation, and Blatt scores will increase significantly as the level of processing of a face judgment varies from "no" to "gender" to "honesty." The semantic quality hypothesis also maintains that recognition RT will decrease significantly as the level of processing of a face judgment varies from "no" to "gender" to "honesty."

The feature quantity hypothesis, on the other hand, predicts that recognition and recognition RT will not vary significantly as a function of the level of processing of a face judgment in this paradigm, because the number of saccades is held constant at zero across all judgment levels. Because processing depth per se is considered not to be the effective determinant of the levels of processing effect, according to the quantity hypothesis, no significant differences in the three task-evoked pupillary measures (mean diameter, peak dilation, and latency to peak dilation) or the Blatt measure are predicted.

Method

Subjects. Eighteen Gettysburg College students (12 women and 6 men) participated.

Stimuli. The stimuli used in Experiment 1 were used in this experiment.

Apparatus. A tachistoscopic shutter (Alfax remote control unit for tachistoscope projector) with 96 (± 5) ms presentation capability was placed in front of the Panasonic stimulus camera during the inspection task. The stimulus camera/shutter fixture was enclosed in a light box so that subject monitor illumination (8 fc as measured by an E G & G radiometer-photometer, model 450-1), was constant at all times—before, during, and after presentation of a stimulus face (room illumination was 3 fc during inspection). Except for these two additions, we used the same apparatus as in Experiment 1.

Design. Both the number of saccades (0) and length of inspection time (96 ms) were constant across all three judgment conditions. As in the first experiment, the single (within-subjects) independent variable was level of processing. Counterbalancing was again used to control stimulus and judgment order effects. For each inspection/judgment trial, we recorded task-evoked pupil data.

We took four dependent measures, two confirming measures and two probe measures.

Confirming measures. The first two measures, recognition and recognition RT, were designed to confirm the (behavioral) replication of the levels of processing effect.

Probe measures. The two remaining measures, task-evoked pupillary response (mean pupil diameter, peak pupillary dilation, and latency to peak pupillary dilation) and Blatt et al. (1981) scores, were obtained, respectively, to probe subjects' cognitive level of processing during face judgment and level of face representation at recognition. Mean peak pupillary dilation and mean latency to peak pupillary dilation were defined as the one data point (out of 94 total) at which peak dilation, and latency to peak dilation, occurred during the inspection/judgment task.

Procedure. The inspection task was the same as in Experiment 1, with the following modification of instructions. For each trial, subjects were told to fixate on the white dot in the center of the monitor and remain fixated for the duration of the trial. "Ready" was announced, followed 1 s later by the brief presentation (96 ms) of the face. Subjects were then given 3 s to make their judgment (in the two experimental conditions), and when the 3 s had passed, "time" was announced (in the control condition no actual judgment was required, but "time" was announced).

The procedure for the recognition task was the same as in Experiment 1.

Results and Discussion

Pupil data were dropped for four subjects. Three subjects were dropped because of monitor illumination problems, and one subject was dropped because a data record was not obtained because of a switch malfunction. For the remaining 14 subjects, six trials (out of 126 total) were lost across five subjects (five trials at the low-processing level and one trial at the high-processing level) because of excessive artifacts (blinks, head and eye movements, etc.) during judgment time. Consequently, three subject means in the low-processing condition and one subject mean in the high-processing condition were based on two trials, and one subject mean for the low-processing condition was based on one trial.

For the remaining 120 trials, small artifacts were dropped, leaving some trials with fewer than 94 data points. Typically, such artifacts lasted 96–160 ms, but there was some variability. Artifacts made up 3.7% of the total 11,280 remaining sampled data points (4.2%, 3.6%, and 3.2% for the no-, low-, and high-processing levels, respectively). Present mean task-evoked pupillary response curves were obtained by averaging the respective 94 data points for each of the 14 subject means at each of the three levels of processing and plotting every other mean ($n = 14$) data point (47 total) at 64 ms intervals.

For seven subject cell means, the mean number of faces correctly recognized was zero (three subject means in both the no- and high-processing conditions and one subject mean in the low-processing condition) so that no recognition RT scores for correct *yes* responses were available for those subjects at those processing levels. The procedure used in Experiment 1 was used to "fill" blank subject cell means so that the repeated measures analysis of variance could be performed for recognition RT.

Confirming measures. Mean hit (recognition) rates, false positive rates, and recognition RT scores are presented in Table 2. There were no significant effects of processing level (face judgment) on hit (recognition) scores, $F < 1$, false positive scores, $F < 1$, or recognition RT scores, $F(2, 34) = 1.52$. These findings indicate that the levels of processing effect was not obtained as a function of processing level (face judgment).

Table 2

Experiment 2: Mean Recognition Performance as a Function of Processing Level (Face Judgment)

Variable	Level of processing		
	No	Low	High
Hit rate	0.54	0.54	0.52
False positive rate	0.32	0.38	0.40
Recognition RT ^a	3.99	3.07	3.94

^a Recognition response time in seconds.

Probe measures. Considering first the pupillary measures, mean pupil diameters for the no-, low-, and high-processing levels were 89.55 ($SD = 13.66$), 89.92 ($SD = 13.12$), and 91.92 ($SD = 16.73$), respectively (these ISCAN pixel scores translate into an actual pupil diameter of approximately 3 mm). An analysis of variance indicated no significant main effect of processing level (face judgment) on pupil diameter, $F < 1$.

Figure 1 shows the mean task-evoked pupillary response for 14 subjects as a function of face judgment during inspection/judgment time. The mean latency to peak pupillary dilations for the no-, low-, and high-processing levels were 1.79 ($SD = .56$), 1.55 ($SD = .47$), and 1.73 ($SD = .51$) s, respectively. An analysis of variance indicated no significant main effect of processing level (face judgment), $F < 1$. A separate analysis of variance between the two experimental conditions also indicated no significant main effect of processing level on latency to peak pupillary dilation, $F(2, 26) = 2.18$, $p > .05$.

The mean peak pupillary dilations for the no-, low-, and high-processing levels were 9.29 ($SD = 5.96$), 8.48 ($SD = 5.23$), and 12.34 ($SD = 6.24$), respectively (these ISCAN pixel dilations translate into actual pupillary dilations of about .31, .28, and .41 mm, respectively). An analysis of variance indicated a significant main effect of processing level (face judgment) on peak pupillary dilation, $F(2, 26) = 4.89$, $p < .02$.

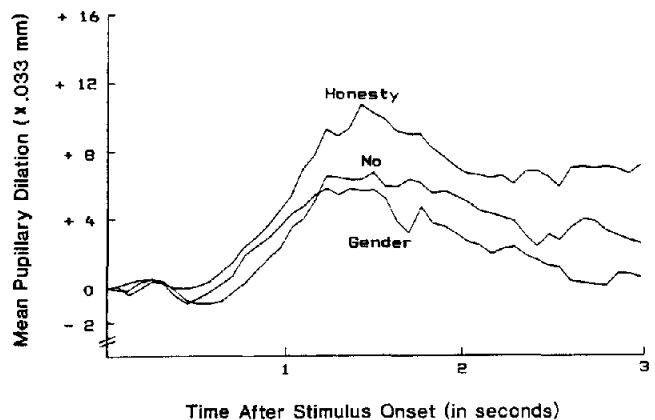


Figure 1. Experiment 2: Mean task-evoked pupillary response for 14 subjects as a function of face judgment during inspection/judgment time (each of the three pupillary response curves is composed of 47 data points, $n = 14$, plotted at 64 ms intervals).

Newman-Keuls tests indicated no significant difference between the no- and low-processing levels, but each of those conditions differed significantly from the high-processing condition (no vs. high, $p < .05$; low vs. high, $p < .05$).

Turning to the second probe measure, the Blatt measure of conceptual level of representation, mean scores for the no-, low-, and high-processing levels were 4.93 ($SD = 2.01$), 4.78 ($SD = 1.94$), and 4.78 ($SD = 2.05$), respectively. An analysis of variance indicated no significant main effect of processing level (face judgment), $F < 1$.

As in Experiment 1, mean pupil diameter and Blatt scores did not vary significantly between the gender and honesty judgment conditions.

The mean peak pupillary dilation measure was found to increase significantly (.13 mm) as level of processing varied from low (gender) to high (honesty).² This finding, taken together with the nonsignificant but somewhat substantial 182-ms increase in the latency to peak pupillary dilation measure across the low (1,554 ms) and high (1,736 ms) processing conditions, provides evidence of differential cognitive processing at the time of initial face inspection. However, even though subjects' cognitive levels of processing (assessed by peak pupillary dilation) increased significantly during inspection and judgment as face judgment varied from gender to honesty, subsequent recognition performance did not improve across those judgment conditions. Presumably, the increased (deeper) processing during inspection did not result in a richer type of memory trace being brought to bear on the subsequent recognition task. Such an interpretation is consistent with the Blatt score findings in both Experiments 1 and 2, which indicate that the level (type) of face representation in memory did not vary significantly as a function of the cognitive level of processing engaged to make a face judgment at initial inspection.

The present peak pupillary dilation and latency to peak pupillary dilation results, obtained in a comparable paradigm, are in agreement with those of Beatty and Wagoner (1978). The present pupillary dilation and recognition findings are consistent with two other studies. Kahneman, Beatty, and Melton (1966), cited in Kahneman and Peavler (1969), found that pupillary dilations during an association learning task were not predictive of later memory. Similarly, Kahneman and Peavler (1969) found that pupillary dilations during an association learning task (within an incentive class) did not predict subsequent recall.

The present inconsistency between the autonomic or physiological (pupillary dilation) measure and the behavioral (recognition performance) measures presents a problem for the semantic quality hypothesis. Although the pupillary measure of cognitive load indicates that extensively greater processing occurred in the honesty judgment condition than in the gender judgment condition during inspection/judgment, such differential processing did not result in a difference in later recognition performance. The nondiffering recognition performance, however, is not a problem for the feature quantity hypothesis because there was no difference between the experimental conditions in number of fixations (i.e., features encoded) at inspection.

The present findings also address obliquely the confounding of eye movements and inspection time, which was the other

consideration raised by the results of Experiment 1. The Experiment 1 findings (that eye movements, inspection time, and subsequent recognition performance increased significantly), together with the Experiment 2 findings (that recognition performance did not vary significantly when eye movements and inspection time were held constant) suggest that it is the number of saccades or the length of inspection time that is associated with the enhanced recognition performance obtained in Experiment 1, the levels of processing effect, rather than processing depth as such. Experiment 3 was designed to specify further the source of the levels of processing effect, particularly as that effect relates to the quantity versus the quality hypotheses under evaluation.

Experiment 3

The purpose of Experiment 3 was to address in a preliminary way the eye movement-inspection time confound of the previous two experiments by isolating the number of eye movements and inspection time. Thus, variability of recognition performance, if obtained, could be clearly attributed to either the number of eye movements or length of viewing time.

One potential approach to isolating the two variables is to equate face inspection time and observe whether the number of eye movements (and subsequent recognition performance) varies significantly. The problem with this approach, of course, is that number of eye movements and length of inspection time are highly correlated. An increase in the number of saccades entails a corresponding increase in viewing time, whereas an increase in viewing time is normally accompanied by an increase in the number of saccades at a rate of 3–4 per s (e.g., Loftus, 1972; Experiment 1).³ It follows

² The fact that peak pupillary dilation in the control condition (no judgment) was comparable to that obtained in the gender judgment condition merits brief consideration. Out of 14 subject means in the control condition, 4 means indicated very little dilation, 5 means indicated moderate dilation (e.g., gender judgment), and 5 means indicated a high amount of dilation (e.g., honesty judgment). Apparently, 10 of the 14 subjects in the control condition did not (at least not consistently) follow instructions but rather made various types of face judgments (e.g., gender, honesty) at the time of initial inspection and judgment.

³ All studies obtaining the levels of processing effect in face recognition that we reviewed have attempted to equate inspection time. In all studies but one, stimuli consisting of photographs made from yearbooks or casting directories (i.e., pictures of head and shoulders) were presented for equal periods of time (e.g., 5 s). Some studies further attempted to equate inspection time by instructing subjects to continue inspecting the slide or picture after making the required judgment. However, because inspection time was contaminated with information other than the face (e.g., neck, clothing, etc.), it appears that only inspection time of each slide or picture was equated. In fact, inspection time of each face was likely to have actually been different (e.g., our Experiment 1). Furthermore, it is likely that number of facial fixations were not equal, resulting in the confounding of eye movements and inspection time (e.g., our Experiment 1).

Some studies (including Wells & Hryciw, 1984, who did present only a face) required subjects to remove their line of sight from the

that, if facial inspection times were equated, number of facial eye movements would also remain comparable. Thus, it does not appear that equating initial face inspection time would definitively isolate number of eye movements from inspection time.

Alternatively, a paradigm in which the number of saccades and inspection time can be isolated experimentally is one in which a stimulus face is presented for the brief interval (175 ms for the average minimum latency found, for example, by Rayner, Slowiaczek, Clifton, & Bertera, 1983) before the eye can execute a saccade. Within the presaccade latency interval, there exists the opportunity to hold equal (at zero) the number of saccades while varying face exposure (inspection) time. We used this paradigm in Experiment 3.

We varied inspection times from a low of 67 ms to a high of 170 ms to meet three criteria. First of all, it was necessary to raise inspection time above the lower threshold (about 20 ms) required for perception and inspection to occur (e.g., see Wever, 1927; Simpson & Crandall, 1972). Second, it was necessary to restrict inspection time to less than 175 ms so that eye movements could not occur. Finally, within that 175-ms saccade latency threshold, we thought it was desirable to establish three exposure conditions to approximate the ratio of inspection durations found between the two experimental conditions in Experiment 1 when inspection time was unrestricted. A comparable duration increment ratio (2.54) within the available 175-ms saccade latency threshold was designed to involve an increase in exposure duration from 67ms to 96 ms to 170 ms.

The question of interest was whether the extra time in the 170-ms inspection time condition, in the absence of additional saccades, would result in significantly improved recognition performance relative to the 96- and 67-ms inspection time conditions. An inspection time hypothesis predicts that as inspection time increases from 67 ms to 96 ms to 170 ms, recognition will increase significantly, recognition RT will decrease significantly, and Blatt scores will not vary significantly. Because the number of saccades is constant at 0, the feature quantity hypothesis predicts that as inspection time increases, recognition, recognition RT, and Blatt scores will not vary significantly.

Method

Subjects. Eighteen Gettysburg College students (10 women and 8 men) participated.

Stimuli. We used the same stimuli as in Experiments 1 and 2 (the nine inspection faces were photocopied and mounted on cardboard, resulting in nine faces of identical size and nearly identical resolution, contrast, etc.).

slide altogether to other areas (e.g., a rating response continuum) during inspection, so that the particular judgment task (e.g., rating of trait[s], physical feature[s], etc.) could be completed. In this latter instance, it seems that inspection time of each slide (not to mention each face) was likely to have been different (again, probably resulting in a difference in the number of eye movements, and thus the eye movement-inspection time confound).

Apparatus. A tachistoscope (Gerbrands Lamp Drive Circuit model 402 A and Gerbrands Integrated Circuit model 300 4-T) was used to present faces to subjects for inspection. The nine stimulus faces were viewed at a distance of 60 cm, which subtended a visual angle of approximately $12^{\circ} 6' \times 15^{\circ} 36'$. During recognition we used the same apparatus as in Experiments 1 and 2.

Design. There were two (within-subjects) independent variables in the present experiment, inspection time and level of processing. Inspection time had three conditions: 67 ms, 96 ms, and 170 ms. Counterbalancing was used to establish the serial presentation order of the three inspection times and the three faces presented for each of the three inspection times. Level of processing was defined as in the first two experiments; counterbalancing was again used to control stimulus and judgment order effects. Each of the three level-of-processing judgments was made for three of the nine faces, and each of the three inspection times was presented for three of the nine faces. In this manner, all three inspection times (67, 96, and 170 ms) were presented at all three processing judgment levels (no, low, and high).

Three dependent measures were obtained, two confirming measures and one probe measure. The first two measures, recognition and recognition RT, were again measured to confirm the (behavioral) replication of the levels of processing effect. The remaining measure, the Blatt et al. (1981) measure, was intended again to probe subjects' level of face representation during recognition and description.

Procedure. The inspection task was the same as Experiment 2 with the following modifications. The inspection task was performed on the tachistoscope. For each trial, we announced "ready," followed 1 s later by the brief presentation (500 ms) of a fixation cross. Subjects were instructed to fixate on the cross immediately. The cross disappeared and the face was briefly presented (67, 96, or 170 ms). Subjects then had 3 s to make their judgment, and when 3 s had passed, "time" was announced.

The procedure for the recognition task was the same as in the first two experiments.

Results and Discussion

Mean Blatt scores of conceptual level of representation are presented in Table 3. There were no significant main effects of inspection time, $F < 1$, or processing level, $F(2, 34) = 1.22$, on Blatt scores; nor was there a significant interaction effect of inspection time and processing level, $F(4, 68) = 1.16$. These Blatt score findings are consistent with those of Experiments 1 and 2.

Mean hit (recognition) rates, false positive rates, and recognition RT scores are presented in Table 4. There were no significant main effects of inspection time or processing level on hit (recognition) rates, $F_s < 1$, or false positive scores, $F_s < 1$, respectively; nor was there a significant interaction effect of inspection time and processing level on hit (recognition)

Table 3
Experiment 3: Mean Conceptual Level of Face Representation as a Function of Inspection Time and Processing Level (Face Judgment)^a

Blatt score	Inspection time (ms)			Level of processing		
	67	96	170	No	Low	High
<i>M</i>	4.59	4.70	4.74	4.89	4.50	4.65
<i>SD</i>	1.86	1.91	1.85	1.89	1.85	1.89

^a As assessed by the Blatt et al. (1981) Conceptual Level of Object Representation measure.

Table 4

Experiment 3: Mean Recognition Performance as a Function of Inspection Time and Processing Level (Face Judgment)

Variable	Inspection time (ms)			Level of Processing		
	67	96	170	No	Low	High
Hit rate	0.59	0.61	0.63	0.56	0.63	0.65
False positive rate	0.38	0.37	0.39	0.35	0.37	0.42
Recognition RT ^a	3.81	3.28	4.02	3.85	4.56	3.44

^a Recognition response time in seconds.

rate, $F < 1$, or false positive scores, $F < 1$. For some subject cell means, no faces were correctly recognized, so that no recognition RT scores for correct *yes* responses were available for those subject cell means (two and one means for the 96- and 170-ms inspection conditions, respectively; and two and one means for the no- and low-processing conditions, respectively). The procedure used in Experiments 1 and 2 was used to fill blank subject cell means, so that the two repeated measure analyses of variance (inspection time and processing level) could be performed for recognition RT. There were no significant main effects of inspection time or processing level on recognition RT scores, $F(2, 34) = .82$ and 2.32 , respectively. These findings indicate that the levels of processing effect in face recognition was not obtained as a function of inspection time or processing level (face judgment).

As inspection time increased by a ratio of 2.5, subsequent recognition increased (4%), but not significantly (likewise recognition RT did not decrease significantly).⁴ This finding is consistent with that of Simpson and Crandall (1972, Experiment 2), who found that as inspection time increased by a ratio of 6.0 (20 ms to 120 ms), "second presentation" face identification findings increased by 9%.

The present findings are consistent with the conclusion that the levels of processing effect in face recognition is not the result of varying (increased) amounts of initial inspection time per se, at least within the range of times evaluated. The data are consistent with the idea that the levels of processing effect in face recognition is attributable to a difference in the number of features viewed and presumably encoded during that time (e.g., Loftus, 1972, 1981, 1983). Such a conclusion is consistent with a model of picture recognition proposed by Loftus (1983), which maintains that recognition performance is the result of the number of features that are encoded.

General Discussion

We designed the present study to shed further light on the semantic quality versus the feature quantity account for the levels of processing effect in face recognition. It is clear that the pattern of results across the three experiments favors the feature quantity hypothesis.

The inference we make from these experiments is that an "encoding specificity principle" (Tulving, 1975; Tulving & Thomson, 1973) operates on what is perceived (facial features)

to determine what is stored. What is stored at facial inspection determines what retrieval cues are effective in accessing that storage at subsequent recognition (Crowder, 1976). In Experiment 1 (when the inspection task was subject-controlled) the differential instructions at the time of inspection induced more eye movements in the honesty judgment condition, which led to the encoding of more features than in the gender judgment condition. Because all features (retrieval cues) were available during the recognition task for both judgment conditions, it may be inferred that the recognition performance difference was due to the encoding difference induced by the initial gender versus honesty judgment instructions. In other words, the facial representation encoded with the honesty judgment at inspection provided more features to be accessed at subsequent recognition than did the gender judgment. Consequently, recognition performance (recognition and recognition RT) improved significantly (e.g., Loftus, 1972, 1981, 1983).

In Experiment 2, the number of saccades was equated at 0 during inspection. Although subjects' cognitive level of processing, assessed by task-evoked peak pupillary dilation, increased significantly as face judgment varied from gender to honesty during inspection and judgment, the deeper processing per se was ineffective ("functionally empty") with respect to subsequent recognition. Because there were no additional fixations during inspection to encode new stimulus features to strengthen the memory network (i.e., the representation of the face), recognition performance, unlike in Experiment 1, did not improve significantly across the gender and honesty judgment conditions.

In Experiment 3, with number of saccades again equated at 0 during inspection, recognition performance did not improve significantly as a function of moderate increases in inspection time, and overall recognition was comparable to that found in Experiment 2.

The definitive experiment that demonstrates the independence of inspection time and number of fixations in the upper time range remains to be conducted. In related work, Loftus (1972, Experiment 2) was afforded the opportunity (because of a large sample, 1,764 trials) to compare picture recognition for observations in which fixations were relatively equal across different inspection periods (e.g., 3 and 5 s), and, conversely, observations in which inspection time was equated (e.g., 3 s) but the number of fixations varied. On the basis of the results of this correlational approach, Loftus concluded that recognition performance is attributable to the number of features encoded during inspection, rather than to length of inspection time per se.

For now, it has been found that in addition to inspection time, a behavioral (eye movement), a physiological (task-evoked pupillary dilation), and a psychometric (Blatt et al., 1981) measure converge predictably on the feature quantity account for the levels of processing effect in face recognition.

⁴ Presumably, the obtained moderate increase in recognition across the three inspection conditions was the result of subjects encoding more information throughout the longer fixation (inspection) period (e.g., see Loftus, 1983).

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